

Forecasting Electricity Prices based on Energy Consumption and Weather

JinYoung Park, Pooria Daneshvar Kakhaki, Silvia Yu

Team 9 | DS5110 Final Project Report

April 20, 2025

1 Summary

This project investigates how electricity prices in Spain are influenced by weather conditions and energy-related factors such as generation and consumption. As electricity markets become more volatile due to climate change and increased renewable integration, understanding the relationship between environmental variables and price is critical for improving decision-making in energy management. Therefore, our goal is to quantify the relationship between environmental variables, energy-related factors, and electricity prices to enable more accurate forecasting. To address this question, we analyzed two primary datasets of 4 years (2015-2018) of hourly data: an energy dataset with data from European Network of Transmission System Operators for Electricity (ENTSOE) and Red Eléctrica de España (REE) – including data on electricity price, total load, and generation by energy source, originally collected for regulatory compliance and market analysis; and a weather dataset from OpenWeather API covering five major Spanish cities – including data on temperature, humidity, wind speed, and weather types (*Hourly Energy Demand Generation and Weather*, 2019). These datasets were specifically selected to help us model how weather patterns and energy production affect pricing structures.

Through Exploratory Data Analysis (EDA) and machine learning modeling such as Linear Regression, Ordinary Least Square (OLS), and Random Forest algorithm, we revealed that Total Load Forecast and Temperature exhibit strong positive correlations with actual electricity prices. Additionally, clear seasonal trends were observed in both demand and temperature distributions. Our best model (Linear Regression) achieved a Root Mean Squared Error (RMSE) of approximately 7.94, indicating a reasonable ability to capture price trends from the available features.

Overall, the project demonstrates that simple models can effectively capture key trends in electricity pricing when powered by well-preprocessed weather and energy features. This analysis provides a foundational approach for developing smarter forecasting tools and supports the ongoing transition to more sustainable and responsive energy systems.

2 Method

2.1 Dataset

As mentioned in the summary, we concentrated on two distinct datasets that were subsequently merged into a single comprehensive one. Initially, the first dataset contained energy data with variables related to generation of energy from different sources, weather elements (related to the energy source like solar and wind) forecasts, electricity load, electricity prices, and temporal information. While the second dataset contained weather-related variables including temperature, pressure, humidity, wind conditions, precipitation (rain and snow), cloud coverage, overall weather condition, and temporal information. Originally, the energy dataset consisted of 29 variables and 35,064 observations, whereas the weather dataset consisted of 17 variables and 178,396 observations. Upon examination, we observed that the energy data contained one hourly observation spanning four years, while the weather data exceeded this by more than fivefold, suggesting potential duplicate observations across the five cities. Further details regarding these variables (including variable names, description, and type) are subsequently provided in Tables A1 and A2 in the Appendix.

2.2 Data Processing

After loading the CSV data files, we started the cleaning of the energy dataset by eliminating columns that were completely missing or contained exclusively zero values. After completing this preliminary cleanup, we proceeded to impute the remaining missing values by implementing a moving average with a window size of 3. Importantly, we selected a moving average approach to ensure appropriate handling of consecutive missing values. By the end of this energy data processing, we had successfully removed insignificant columns, replaced missing data, and decreased the number of variables from 29 to 21.

Subsequently, we advanced to processing the weather data using a similar cleaning approach, with additionally removing irrelevant columns and condensing redundant weather-related categorical variables through one-hot encoding. More specifically, to comprehend the relationship between the two weather-related categorical variables, we initially created a Sankey plot (as illustrated in Figure A1 in the Appendix) mapping the broader Weather Main values to more specific Weather Description values. This visualization showed many values with only minimal observations recorded. Consequently, we decided to consolidate the weather conditions into five primary categories – clear (clear conditions), cloudy (cloud cover), precipitation (rain, drizzle, thunderstorm, and squall), low visibility (mist, fog, smoke, dust), and snow (snow conditions). Following this consolidation, we converted the Weather Main variable into five one-hot encoded features and eliminated the original two weather-related categorical variables. As previously mentioned, the datasets contained differing amounts of observations, which necessitated condensing the weather data by aggregating observations by timestamp, averaging data across cities for the remaining variables, and eliminating duplicate observations. Upon the completion of the weather data processing, we reduced the number of observations from 178,396 to 35,064, while maintaining the same number of variables despite changing their nature.

Finally, we completed our data preparation by merging the two cleaned datasets through an inner join operation on the datetime field, yielding a consolidated dataset that encompassed both energy and weather factors. The resulting merged dataset contained 37 variables and 35,064 observations; additional details regarding these variables are subsequently provided in Table A3 in the Appendix.

2.3 Modeling

After processing and merging the datasets, we performed exploratory data analysis to assess the relationship between features and the target variable – price actual. We produced histograms of each feature, line plots of each feature over time, and scatter plots of each feature versus the target variable. These visualizations enabled us to identify potential patterns and correlations within the data before proceeding to model development.

Following the feature analysis, we further processed the data by separating the variables, splitting the dataset, and standardizing the features. First, we isolated the features, the target variable, and timestamps as separate components. Next, we divided the dataset into an 80% training and 20% test set, while maintaining chronological order. Specifically, the training set comprised the initial 80% of the 4-year time series, while the test set contained the subsequent 20%. Finally, we standardized the features using StandardScaler from the scikit-learn library in Python to ensure all features are on the same scale and the impact of outliers are reduced.

Next we trained multiple models to predict electricity prices, beginning with simpler linear approaches and progressing to more complex tree-based algorithms. Initially, we used the Python package

statsmodels to perform the Ordinary Least Square (OLS) by adding constant terms to both the features and the response variable before fitting the model with the training set and constants. We then implemented a standard Linear Regression model by importing and calling the LinearRegression() function and fitted the model with the same training data to establish a comparison baseline.

Subsequently, to address potential overfitting, we trained Ridge and Lasso Regression models with cross-validation over logarithmically spaced regularization strengths from 10^{-8} to 10^8 , ensuring optimal parameter selection. The Lasso Regression implementation included an additional 5-fold cross-validation parameter to further refine the selection process.

Following these regularization linear models, we implemented three advanced algorithms, each incorporating hyperparameter optimization techniques to enhance predictive accuracy and mitigate overfitting. Through comprehensive grid search procedures with cross-validation (5-fold cross-validation using negative MSE as scoring metric), we systematically identified the optimal configuration for each model's unique set of parameters. This optimization process ensured that the final implementations represented the most effective versions possible for our specific electricity price prediction task. The advanced models include Decision Tree Regression with grid search over hyperparameters including max_depth (3, 5, 10, 20, 50, None) and min_samples_split (2, 5, 10, 15, 20), Random Forest Regression with grid search over Hyperparameter optimization for n_estimators (20, 50, 100), max_depth (None, 3, 5, 10, 15), and min_samples_split (2, 5, 10), and finally XGBoost Regressor with grid search parameter tuning for n_estimators (50, 100), max_depth (3, 6, 10), learning_rate (0.01, 0.1, 0.3), and subsample (0.8, 1.0).

Upon completion of the model training, we evaluated each model's performance by calculating the R-squared values and Mean Squared Error (MSE) on the training and test sets. These metrics provided a quantitative basis for comparing predictive accuracy across our spectrum of modeling approaches.

2.4 Model Evaluation

After training the models, we used the MSE on the test set for each model to assess generalization performance. For the best performing model, we conducted validations to assess its predictive accuracy and statistical properties. The model's performance was first evaluated through the time series visualization, where we plotted the actual and predicted prices over time for both the training and test datasets. This plot allowed us to visualize how well the model captured the price fluctuations over time. Then to better understand the underlying trends without the noise, we also smoothed the plot using a 32-point rolling average window. Next, to identify any systematic prediction errors and potential heteroscedasticity in the data, we performed residual analysis by plotting the difference between actual and predicted values over time. Then to assess the normality assumption of the residuals, we generated a QQ plot comparing the empirical distribution of residuals and theoretical normal distribution. Lastly, we assessed the model summary to identify the significant predictors (by evaluating the p-values and t-statistics of all features) and examine the MSE and Root Mean Squared Error (RMSE) values.

3 Results

This section presents the key findings from our exploratory and predictive modeling efforts. The focus is on evaluating the performance of the linear regression model and visualizing the model's behavior in comparison with the actual data.

3.1 Model Performance

As described in the Methods section, we trained multiple models to predict electricity prices. Here, we report the MSE on the test set for each model to assess generalization performance. The results are summarized in Table 1:

Table 1: Mean Squared Error (MSE) on the Test Set

Model	MSE
OLS	63.5404
Linear Regression	63.5404
Ridge Regression	63.5651
Lasso Regression	87.5309
Decision Tree	81.6103
Random Forest	114.6185
XGBoost	90.0091

- Linear Regression and OLS yielded identical performance, as expected, since they are equivalent under default settings. Their MSE of 63.54 indicates a reasonable baseline fit.
- Ridge Regression slightly outperformed Linear Regression in training but had a nearly identical MSE on the test set, suggesting minimal regularization benefit for this feature set.
- Lasso Regression performed significantly worse, with an MSE of 87.53. This suggests that the strong L1 penalty may have removed important predictive features, leading to underfitting.
- Decision Tree and XGBoost showed moderate performance but failed to outperform linear models.
- Random Forest, unexpectedly, had the highest MSE (114.62). This could indicate overfitting on the training data or poor hyperparameter choices.

We performed cross-validation with grid search to select the best hyperparameters for each model, aiming to achieve optimal performance and reduce overfitting. Despite this effort, the more complex models—such as Random Forest and XGBoost—still failed to outperform the simpler linear models. This suggests that the underlying relationships in the data may be 1) largely linear, 2) not expressive enough to benefit from the capacity of non-linear models. Our earlier analysis indicated a strong correlation between the actual electricity price (the target) and the day-ahead forecasted price. It is possible that linear models effectively capture this relationship, but complex models overfit to the features, which reduce their generalization performance.

3.2 Predicted vs. Actual Prices

To assess the model's performance qualitatively, we plotted the actual and predicted prices over time for both the training and test datasets.

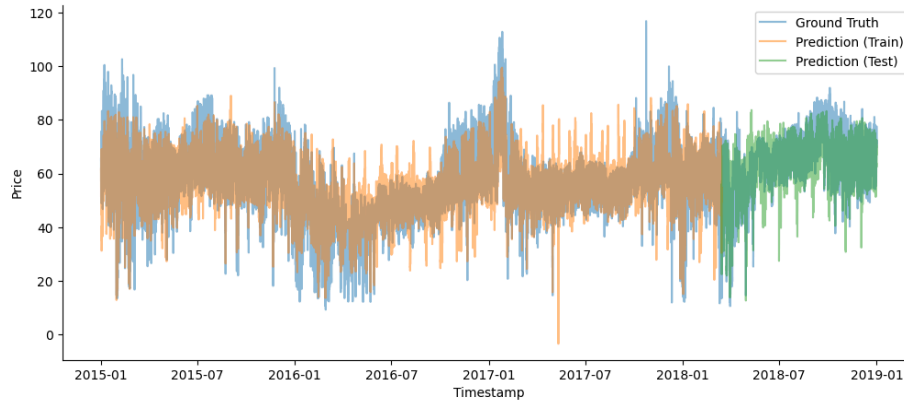


Figure 1. Time series of actual versus predicted electricity prices for a randomly selected 500-hour window.

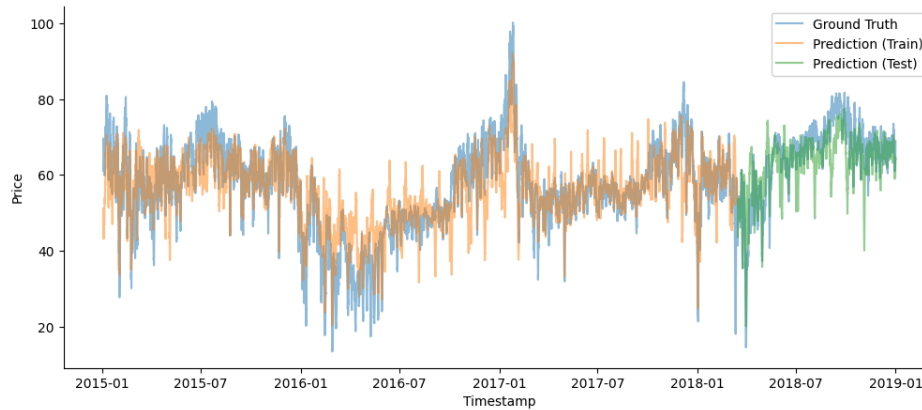


Figure 2. Smoothed time series of actual versus predicted electricity prices using 32-point rolling average.

These plots demonstrate that the model captures general price fluctuations but sometimes fails to match the magnitude of extreme price spikes. For example, the decrease of the electricity price in the middle of 2016 is reflected in the predictions of the training set, however the magnitude does not match the ground truth, as seen in Figure 2 where the orange does not perfectly match the blue lines. The overall upward trend in the test set is also captured by the model.

3.3 Residual Analysis

Residuals, defined as the difference between actual and predicted prices, help identify model deficiencies.

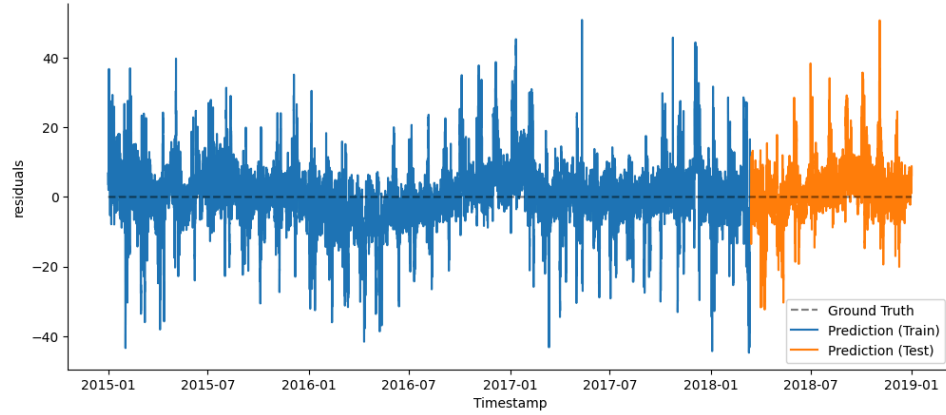


Figure 3. Residuals over time for both training and test sets.

Although residuals are centered around zero, we observe patterns in the residuals. This suggests that the target variable is not fully captured by the model, and temporal models (e.g., Transformers, LSTM) may yield better performance.

3.4 QQ Plot

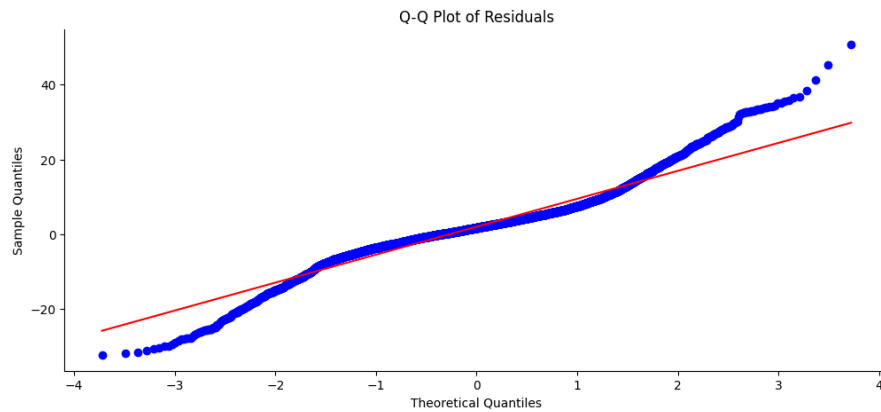


Figure 4. Quantile-Quantile (QQ) plot of residuals.

The QQ plot shows moderate deviations from the theoretical normal line, particularly in the tails, indicating non-Gaussian residuals.

3.5 OLS Summary

A summary of the Ordinary Least Squares (OLS) regression reveals:

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                        OLS Regression Results
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Dep. Variable:          price actual    R-squared:                0.605
Model:                  OLS            Adj. R-squared:           0.604
Method:                 Least Squares   F-statistic:              1259.
Date:                   Sat, 19 Apr 2025 Prob (F-statistic):       0.00
Time:                   18:44:28        Log-Likelihood:          -1.0123e+05
No. Observations:      28036           AIC:                    2.025e+05
Df Residuals:          28001           BIC:                    2.028e+05
Df Model:               34
Covariance Type:       nonrobust
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	coef	std err	t	P> t	[0.025	0.975]
const	-0.8706	3.272	-0.266	0.790	-7.285	5.543
generation biomass	0.0122	0.001	11.254	0.000	0.010	0.014
generation fossil brown coal/lignite	-9.118e-05	0.000	-0.348	0.728	-0.001	0.000
generation fossil gas	0.0001	4.76e-05	2.400	0.016	2.1e-05	0.000
generation fossil hard coal	-0.0003	6.57e-05	-4.006	0.000	-0.000	-0.000
generation fossil oil	0.0014	0.001	0.978	0.328	-0.001	0.004
generation hydro pumped storage consumption	0.0005	0.000	4.580	0.000	0.000	0.001
generation hydro run-of-river and poundage	0.0006	0.000	2.134	0.033	4.52e-05	0.001
generation hydro water reservoir	0.0003	6.33e-05	5.411	0.000	0.000	0.000
generation nuclear	6.732e-05	7.44e-05	0.905	0.366	-7.85e-05	0.000
generation other	-0.0127	0.004	-3.567	0.000	-0.020	-0.006
generation other renewable	-0.0241	0.007	-3.379	0.001	-0.038	-0.010
generation solar	0.0012	0.000	4.185	0.000	0.001	0.002
generation waste	-0.0242	0.002	-14.555	0.000	-0.027	-0.021
generation wind onshore	-0.0004	0.000	-1.622	0.105	-0.001	8.57e-05
forecast solar day ahead	5.299e-05	0.000	0.179	0.858	-0.001	0.001
forecast wind onshore day ahead	0.0013	0.000	4.927	0.000	0.001	0.002
total load forecast	-5.814e-05	0.000	-0.507	0.612	-0.000	0.000
total load actual	0.0002	0.000	1.881	0.060	-9.57e-06	0.000
price day ahead	0.7828	0.008	98.302	0.000	0.767	0.798
temp	2.3515	0.235	9.991	0.000	1.890	2.813
temp_min	-1.1225	0.129	-8.708	0.000	-1.375	-0.870
temp_max	-1.1836	0.113	-10.505	0.000	-1.404	-0.963
pressure	-3.779e-06	1.78e-05	-0.212	0.832	-3.87e-05	3.12e-05
humidity	0.0723	0.007	10.548	0.000	0.059	0.086
wind_speed	-2.0481	0.053	-38.802	0.000	-2.152	-1.945
wind_deg	-0.0085	0.001	-8.197	0.000	-0.010	-0.006
rain_1h	-0.3011	0.310	-0.971	0.332	-0.909	0.307
rain_3h	-39.2851	14.461	-2.717	0.007	-67.629	-10.941
snow_3h	3.4924	0.483	7.224	0.000	2.545	4.440
clouds_all	-0.0165	0.006	-2.621	0.009	-0.029	-0.004
clear	0.4514	0.942	0.479	0.632	-1.395	2.298
cloudy	-1.3909	0.927	-1.501	0.133	-3.207	0.425
low_visibility	6.8720	1.031	6.664	0.000	4.851	8.893
precipitation	-1.4667	1.005	-1.459	0.145	-3.437	0.504
snow	-5.3364	2.772	-1.925	0.054	-10.769	0.096

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Omnibus:                1918.247    Durbin-Watson:           0.126
Prob(Omnibus):          0.000      Jarque-Bera (JB):        9522.961
Skew:                   -0.088      Prob(JB):                0.00
Kurtosis:               5.850       Cond. No.                2.47e+15
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Significant predictors include Price Day Ahead, Generation from Solar, Biomass, and Waste (from energy-related attributes), as well as Temperature, Humidity, Wind Degree, and Wind Speed (from weather-related attributes). All of these variables have p-values < 0.001 and high t-statistics, indicating strong statistical significance in the OLS model.

An MSE of 63.54 corresponds to an RMSE of approximately 7.94, which represents the standard deviation of the model's prediction errors. While not a direct measure of average error like MAE, the RMSE indicates that most predictions fall within roughly 8 euros per megawatt-hour of the actual price, with larger errors penalized more heavily.

4 Discussion

This project set out to forecast electricity prices in Spain using a combination of energy-related variables and weather data. Our goal was to understand how environmental and consumption patterns influence electricity pricing and to develop predictive models that can support better energy planning and sustainability. To achieve our goal, we merged and cleaned datasets of energy and weather information,

conducted exploratory data analysis, and applied machine learning models to evaluate predictive performance.

The results show that electricity pricing is influenced by both energy demand and weather conditions. The Mean Squared Error (MSE) for our best-performing model was 63.54, and our model was able to capture some of the trends in the fluctuations of the electricity price.

4.1 Future Recommendations

To build on these findings and address current model limitations, we suggest the two following future directions. First, our current models treat each time point independently, ignoring the temporal dependencies inherent in electricity pricing. Incorporating time-series models (e.g., ARIMA, LSTM, or Transformer-based models) could better capture trends, seasonality, and delayed effects of demand or weather changes. Including a window of prior time points as input features may help model transitions and sudden shifts in prices. This approach could be especially useful in capturing peak demand patterns or delayed responses to environmental changes. Second, the current model aggregates weather data across major cities. Developing city-level models would allow us to account for regional differences in consumption, weather patterns, and generation capabilities, ultimately improving the model's spatial resolution and accuracy.

5 Statement of Contribution

Our team worked both independently and collaboratively throughout the project. We initially processed data separately, then met to compare results. For example, we made collective decisions about column removal and weather dataset condensation. JinYoung Park focused primarily on exploratory data analysis and helped interpret features. Pooria Daneshvar Kakhaki focused primarily on model development and studying weather data. Silvia Yu focused primarily on data processing and contributed to the implementation of linear regression and OLS models.

6 Reference

1. *Hourly energy demand generation and weather*. (2019, October 10). Kaggle.
<https://www.kaggle.com/datasets/nicholasjhana/energy-consumption-generation-prices-and-weather/data>

7 Appendix

Weather Main → Weather Description (No Loops)

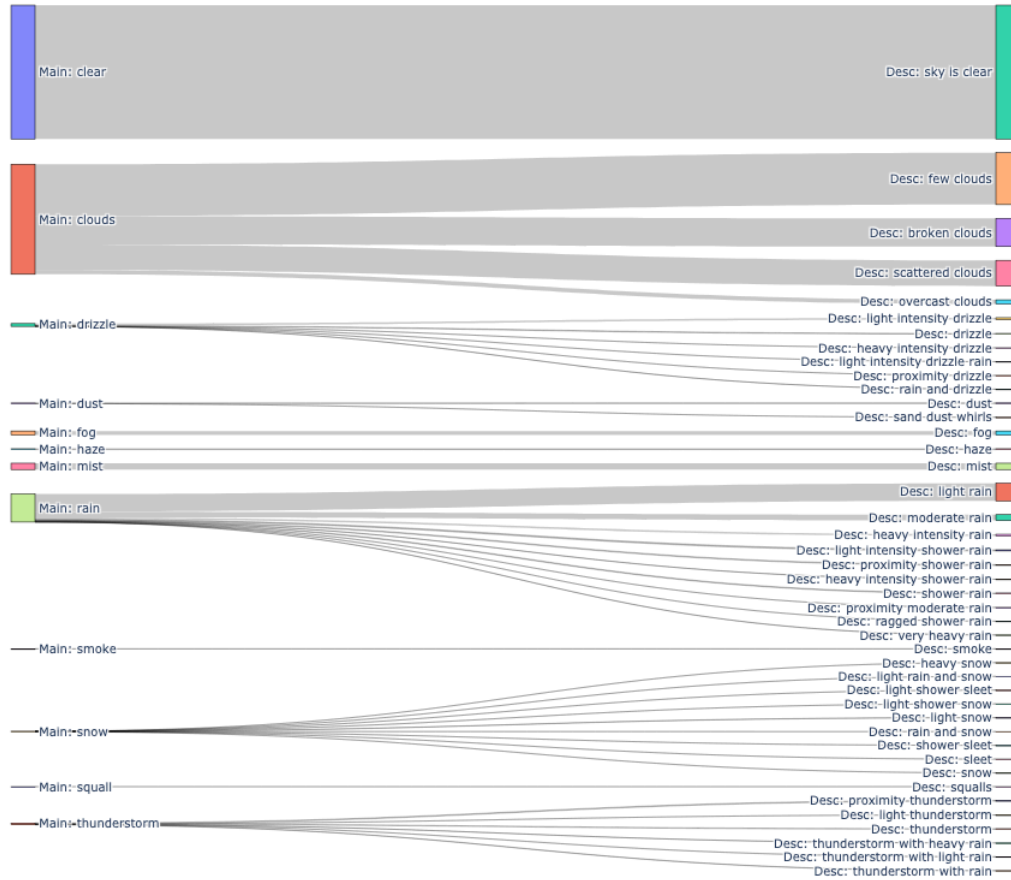


Figure A1. Sankey plot of the Weather Main and Weather Description variables.

Table A1. Summary of the variables in the original energy dataset.

Variable	Description	Type
Time	Year month day time in CET	Continuous
Generation Biomass	Megawatt	Continuous
Generation Fossil Brown Coal/Lignite	Megawatt	Continuous
Generation Fossil Coal-derived Gas ²	Megawatt	Continuous
Generation Fossil Gas	Megawatt	Continuous
Generation Fossil Hard Coal	Megawatt	Continuous

Generation Fossil Oil	Megawatt	Continuous
Generation Fossil Oil Shale ²	Megawatt	Continuous
Generation Fossil Peat ²	Megawatt	Continuous
Generation Geothermal ²	Megawatt	Continuous
Generation Hydro Pumped Storage Aggregated ¹	Megawatt	Continuous
Generation Hydro Pumped Storage Aggregated	Megawatt	Continuous
Generation Hydro Run-of-River and Poundage	Megawatt	Continuous
Generation Hydro Water Reservoir	Megawatt	Continuous
Generation Marine ²	Megawatt	Continuous
Generation Nuclear	Megawatt	Continuous
Generation Other	Megawatt	Continuous
Generation Other Renewable	Megawatt	Continuous
Generation Solar	Megawatt	Continuous
Generation Waste	Megawatt	Continuous
Generation Wind Offshore ²	Megawatt	Continuous
Generation Wind Onshore	Megawatt	Continuous
Forecast Solar Day Ahead	Megawatt	Continuous
Forecast Wind Offshore Day Ahead ¹	Megawatt	Continuous
Forecast Wind Onshore Day Ahead	Megawatt	Continuous
Total Load Forecast	Megawatt	Continuous
Total Load Actual	Megawatt	Continuous
Price Day Ahead	Euro per Megawatt Hours	Continuous
Price Actual	Euro per Megawatt Hours	Continuous

¹Column was dropped because all data values were missing.

²Column was dropped because all data values were zeros.

Table A2. Summary of the variables in the original weather dataset.

Variable	Description	Type
Time	Year month day time in CET	Continuous
City Name ³	Barcelona/ Bilbao/ Madrid/ Seville/ Valencia	Nominal
Temperature	Kelvin	Continuous
Temperature Minimum	Kelvin	Continuous
Temperature Maximum	Kelvin	Continuous
Pressure	Hectopascal (hPa)	Continuous
Humidity	Percent	Continuous
Wind Speed	Meter per second	Continuous
Wind Degree	Wind Direction	Continuous
Rain 1h	Rain in the last hour in millimeters	Continuous
Rain 3h	Rain in the last 3 hour in millimeters	Continuous
Snow 3h	Snow in the last 3 hours in millimeters	Continuous
Clouds All	Percent of cloudiness	Continuous
Weather Id ³	Code used to describe weather	Nominal
Weather Main ⁴	Rain/Snow/Clouds/Etc.	Nominal
Weather Description ³	Long description	Nominal
Weather Icon ³	Weather icon id for the website	Nominal

³Column was dropped after processing

⁴Column was split into 5 columns after one-hot encoding

Table A3. Summary of the variables in the merged dataset.

Variable	Description	Type
Time	Year month day time in CET	Continuous

Generation Biomass	Megawatt	Continuous
Generation Fossil Brown Coal/Lignite	Megawatt	Continuous
Generation Fossil Gas	Megawatt	Continuous
Generation Fossil Hard Coal	Megawatt	Continuous
Generation Fossil Oil	Megawatt	Continuous
Generation Hydro Pumped Storage Aggregated	Megawatt	Continuous
Generation Hydro Run-of-River and Poundage	Megawatt	Continuous
Generation Hydro Water Reservoir	Megawatt	Continuous
Generation Nuclear	Megawatt	Continuous
Generation Other	Megawatt	Continuous
Generation Other Renewable	Megawatt	Continuous
Generation Solar	Megawatt	Continuous
Generation Waste	Megawatt	Continuous
Generation Wind Onshore	Megawatt	Continuous
Forecast Solar Day Ahead	Megawatt	Continuous
Forecast Wind Onshore Day Ahead	Megawatt	Continuous
Total Load Forecast	Megawatt	Continuous
Total Load Actual	Megawatt	Continuous
Price Day Ahead	Euro per Megawatt Hours	Continuous
Price Actual	Euro per Megawatt Hours	Continuous
Temperature	Kelvin	Continuous
Temperature Minimum	Kelvin	Continuous
Temperature Maximum	Kelvin	Continuous
Pressure	Hectopascal (hPa)	Continuous

Humidity	Percent	Continuous
Wind Speed	Meter per second	Continuous
Wind Degree	Wind Direction	Continuous
Rain 1h	Rain in the last hour in millimeters	Continuous
Rain 3h	Rain in the last 3 hour in millimeters	Continuous
Snow 3h	Snow in the last 3 hours in millimeters	Continuous
Clouds All	Percent of cloudiness	Continuous
Clear ⁵	Main weather condition	Discrete
Cloudy ⁵	Main weather condition	Discrete
Low visibility ⁵	Main weather condition	Discrete
Precipitation ⁵	Main weather condition	Discrete
Snow ⁵	Main weather condition	Discrete

⁵Column resulting from one-hot encoding of the “Weather Main” variable.